

Srinivasulu D

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Experienced Embedded Senior Engineer with around 7 years of experience in Embedded Android Platforms, Automotive Infotainment Systems, Linux Kernel and Device Driver Development, and IoT Solutions. Skilled in C/C++, Linux internals, BSP, Yocto, and board bring-up. Strong analytical and debugging skills with proven expertise in performance optimization, system integration, and end-to-end embedded product development.

Technical Skills

- Proficient in C/C++ programming, data structures and OS fundamentals
- Skilled in Linux device drivers, kernel internals, board bring-up and system debug.
- Skilled in SoC profiling, modelling, benchmark estimation and analysis.
- Understanding of ARMv7/v8 architecture, kernel scheduler and memory management.
- Experienced with GIT, Gerrit and perforce.
- Code navigation tools like Source insight, Open Grok, c-tags and c-scope.
- Experienced in profiling and analysis tools like Perfetto, trace-events.
- Strong with debugging tools like Lauterbach Trace-32, JTAG, crash utility, and Debug Insight.
- Awareness of JIRA for issue tracking and Confluence for documentation.
- Experience in Linux build systems; Tools: GNU Toolchain, Yocto-Build root.
- Have good understanding in U-Boot, Build Root Device Tree Customizations
- Having good Experience in using tools like Gdb, Val grind, Wireshark, KGdb
- Experience in android multimedia framework, have worked on android audio split use case for android automotive.
- Exposure in development of automotive infotainment (Rear-seat entertainment) systems.
- Enabled Multi Display support in Android9 by back porting of Audio Multi-zone feature from Android10. for both QC820 Automotive platform and Hikey960 Mobile platform.
- Development of Multimedia Android Applications and building on different Android flavors for different platforms.
- Worked on Android Porting, Android Customization and on Android Build structures.

Professional Experience

Harman India Pvt Ltd, Bangalore

MAY 2022 - PRESENT

Embedded Senior Engineer

Project 01: Embedded Linux BSP Development, Secure Boot Enablement, and WP7702 Integration on Microchip SAM5D3 Platform

Project Summary: This project involved complete development of an embedded Linux platform from scratch for the Microchip SAM5D3-based processor. The goal was to design and bring up a secure, stable, and testable Linux system with modem connectivity and automated hardware testing capability. The development included secure boot enablement, BSP setup, kernel and driver development, and WP7702 modem integration. A comprehensive test application was created to validate multiple hardware interfaces and drivers under varying environmental conditions through chamber testing.

Roles and Responsibilities:

- Developed the complete BSP (Bootloader, Kernel, Root Filesystem) from scratch for the SAM5D3 platform using Yocto Kirkstone (4.0).
- Enabled Secure Boot by generating signed bootloader, FIT kernel images, and enforcing cryptographic validation during boot.

- Integrated WP7702 modem for LTE connectivity by porting drivers, updating device tree nodes, and validating AT and data communication interfaces.
- Implemented OTA update support using Mender/SWUpdate for remote firmware upgrade and rollback functionality.
- Conducted system stability, performance, and regression testing after secure boot and OTA integration.
- Carried out root cause analysis for kernel panics, watchdog resets, and hardware communication errors.
- Developed a C-based hardware test application to verify drivers and system health — including SPI, I2C, GPIO, UART, ADC, and Touch interfaces.
- Performed chamber testing under different temperature conditions to validate hardware stability and driver reliability.
- Contributed to CI/CD test pipelines, maintaining automated test scripts and unit test verification for kernel modules.

Project 02: Unified memory Reclaim / System Memory

Project Summary: Improved efficiency of structure to secure Android Memory with Linux kernel- based Memory management Solution. Developing a unified memory load determination criteria and reclaim action execution for advanced memory management, resulting in clear improvement of memory structure to secure memory as much as necessary in an optimal moment with minimal resources, improved memory policy management including huge page configuration, and Damon based memory observability for reclaim and memory configuration management. Verifying the developed unified policy and applying it to commercialization project.

Roles and Responsibilities:

- Damon (Data Access Monitor) module porting across chipset & stabilizing.
- Memory Workload analysis for targeted solution development & tools development for memory workload analysis.
- Kernel stability assurance & commercialization support across chipset.
- Kernel mainline feature porting & Evaluation in Samsung devices.
- Arm64 ASM optimized kernel compression algorithms.
- Test suite development for project KPI: SPAT Test (Sluggish Performance Analysis Test).

Project 03: Samsung SmartThings

Project Summary: Involved in developing and maintaining features on TizenRT, the RTOS used for SmartThings IoT devices.

Roles and Responsibilities:

- Worked on the TizenRT OS. Involved in the debugging, configuration changes and testing.
- Work on the changes in the linux image, device-tree, kernel drivers based on the requirements. Making kernel patches, and integration of the linux image changes.

HCL Technologies Ltd. (R&D), Bangalore

NOV 2020 – MAY 2022

Software Engineer

Project 04: iMX based platform for Vehicle infotainment cluster.

Project Summary: This project is the Involved in bringing up the platform for digital cockpit product Which is Android based infotainment system.

Roles and Responsibilities:

- Customize/Configure/fix-issues in Linux kernel modules/features for automotive platform based out iMX platforms.
- Board brings up and porting to newer hardware platforms according to Client requirements
- Development and stabilization of Linux device drivers for hardware peripherals/interfaces (UART, SPI, I2C, GPIO, USB)

- Debugged kernel panics, deadlocks, memory leaks, memory/stack corruptions, system hangs, bitflips, watchdog resets, voltage margin and more.
- Handled build, compilation and OTA-related issues on Android.
- Developed automation scripts to simplify tasks, enhancing efficiency and productivity.
- Supported multiple NXP SoCs based on 6th/7th/8th generation platforms and NXP chipsets.

Sasken Technologies Ltd. (R&D), Pune

MAR 2019 – SEPT 2020

Embedded Software Engineer

Project 05: Android IVI Multimedia Audio Routing

Project Summary: Main purpose of this project is that a user should be able to simultaneously play two separate audio/multimedia streams and respectively route individual audio to two separate audio playback devices (one on BT while other on headset/speaker). As Multi Display support feature is already added in Android10 but not in Android9 so understand the workflow of Android10 audio routing and backported to Android9 to enable Multi Display support.

Roles and Responsibilities:

- Understanding of multi-display feature in Android 10 and Android Audio subsystem.
- Back porting the feature on different platforms Hikey960 Mobile platform and QC820 and 8155 Automotive platform.
- Involving in feature discussions and interacting with remote teams.
- Involved in fixing the reported bugs during the testing cycle.
- Involved in fixing code analysis tools bugs in Settings and control modules.
- Understanding ALSA driver flow.

Education

Post-Graduation Diploma: Embedded System Design

JUL 2018 – FEB 2019

Vector India - Chennai

Bachelor's Degree: Electrical and Electronics

AUG 2015 – JUN 2018

Jawaharlal Nehru Technological University - Anantapur

Diploma: Electrical and Electronics

AUG 2012 – JUN 2015

VRS& YRN Group of Educational Institutions - Chirala

Achievements

- Received SPOT award for successfully bringing up Microchip SAMA5D3.
- Received recognition for excellent contribution to 5GVPN library for Linux Platform.
- Received recognition for support to Multizone Audio for Qualcomm chipsets.

Core Strengths

- Diligent • Articulate • Adaptable and Punctual • Analytical • Inquisitive • Receptive
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